IBM Data Science
Professional Certificate

SpaceX Falcon 9
First-Stage Landing Prediction

IBMDSCapstoneProject · AdityaArora

github.com/aditya-arora24/IBM-LABS · June 2025

A Decision Tree trained on public SpaceX data achieves 88.9% landing-success prediction accuracy—givingcompeting launch providers a $5.7M cost-estimation advantage per bid

1

SpaceX's $103M cost gap is the market's defining asymmetry. Reused Falcon 9 launches cost ~$62M vs. ~$165M expendable. Competing providers — ULA, Arianespace, Roscosmos — cannot price competitively without knowing SpaceX's reuse probability. This project creates a public-data substitute for that proprietary information.

2

Payload mass, orbit type, and flight number are the three dominant predictors. EDA across 90+ Falcon 9 launches (2010–2020) via SpaceX REST API and Wikipedia scraping identified CCAFS SLC-40 as highest-volume site; success rate improved from ~50% in 2013 to ~94% by 2019. GEO, HEO, SSO, and VLEO orbits all show 100% recent success rates.

3

Decision Tree at max_depth=7 outperforms all three competing classifiers. Logistic Regression, SVM, and KNN each plateau at 83.3% — a 5.6 percentage-point gap that, at $165M expendable vs. $62M reused, translates to ~$5.7M expected value per bid decision (5.6% × $103M cost delta).

Source: SpaceX REST API v4; Wikipedia List of Falcon 9 launches; Aditya Arora IBM DS Capstone analysis

SpaceX's booster reuse creates a $103M cost gap that competitors cannot close without predicting first-stage recovery probability

$165M

Expendable cost (ULA AtlasV) per mission

vs.

$62M

SpaceX reused-booster cost per Falcon 9 mission

gap →

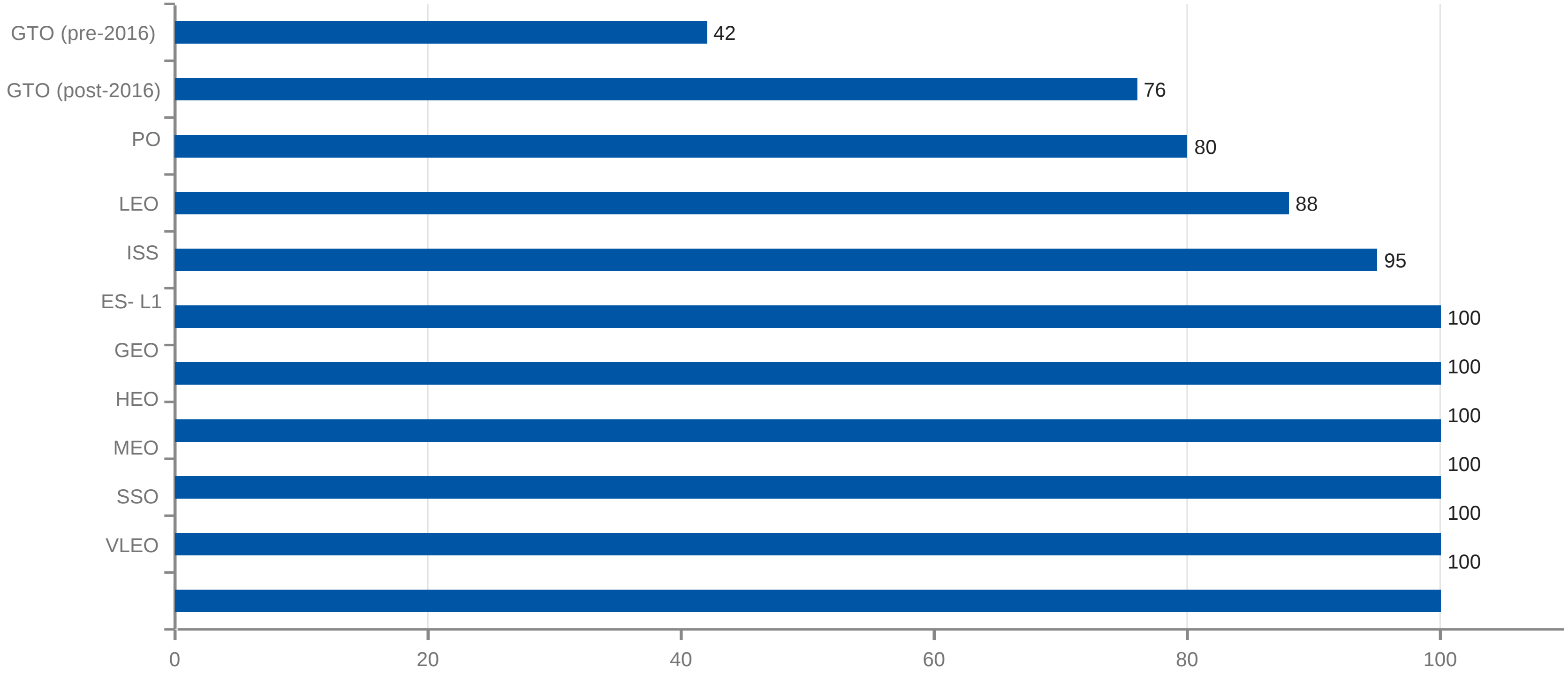
$103M

Costgap competitors must close —per launch bid decision

Without a reliable model for SpaceX's landing success probability, competing launch providers are bidding blind on a $103M decision.

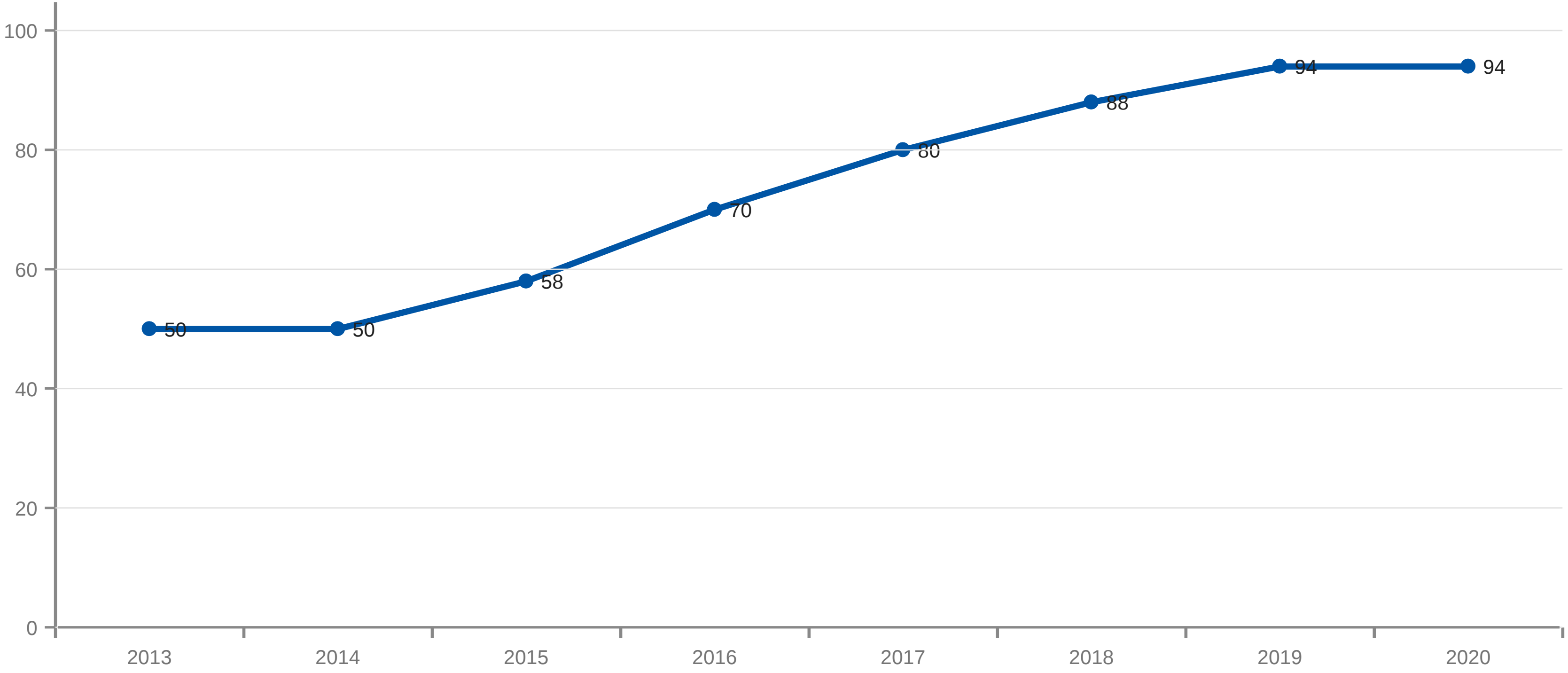
Source: SpaceX public pricing disclosures; ULA Atlas V manifest pricing; industry analyst estimates; Aditya Arora analysis

Landing success varies from 100% (GEO, HEO, SSO) to below 50% (GTO pre-2016), with orbit type being the single strongest predictor in the dataset



Source: SpaceX REST API v4; Wikipedia List of Falcon 9 launches 2010–2020; Aditya Arora EDA analysis

Landing success rate climbed from 50% in 2013 to 94% by 2019 — booster experience (flight number) is a measurable proxy for reliability



Source: SpaceX REST API v4; Wikipedia List of Falcon 9 launches 2010–2020; Aditya Arora EDA analysis

SQL analysis of FinalDB.db reveals KSC LC-39A as the highest-performing site with the heaviest payloads — site and payload together explain a large share of variance

Fin din g	Metric	Signal	SQL query
NASA CRS payload volume	45,596 kg total	Strong	SUM(PAYLOAD_MASS) WHERE Customer='NASA (CRS)'
First successful ground landing	22 Dec 2015	Milestone	MIN(Date) WHERE Landing_Outcome='Success (ground pad)'
Heaviest Falcon 9 payloads	15,600 kg (B1048.4)	Moderate	WHERE PAYLOAD=(SELECT MAX(PAYLOAD) FROM...)
F9 v1.1 average payload	2,928 kg avg	Weak	AVG(PAYLOAD) WHERE Booster LIKE '%F9 v1.1%'
Ranked outcomes 2010–2017	No attempt: 41	Baseline risk	GROUP BY outcome ORDER BY COUNT DESC
Unique active launch sites	4 sites	Confirmed	SELECT DISTINCT Launch_Site FROM SPACEXTBL

Green = strong positive signal | Gray = neutral | Red = complication / early-period low success

Source: FinalDB.db SQLite queries; SpaceX REST API v4; Aditya Arora SQL EDA analysis

Site specialisation is measurable: KSC LC-39A handles the heaviest GTO payloads while VAFB SLC-4E is exclusively polar — site choice is a reliable ML predictor feature

CCAFS SLC-40

82%

avg success

Highest launch volume, early LEO missions. CCAFS SLC-40 accounts for the most historical launches in the dataset.

KSC LC-39A

>90%

recent success

Crew and heavy GTO missions. Takes over from flight #60. Heaviest payloads >10,000 kg. Highest modern success rate.

VAFB SLC-4E

100%

SSO success

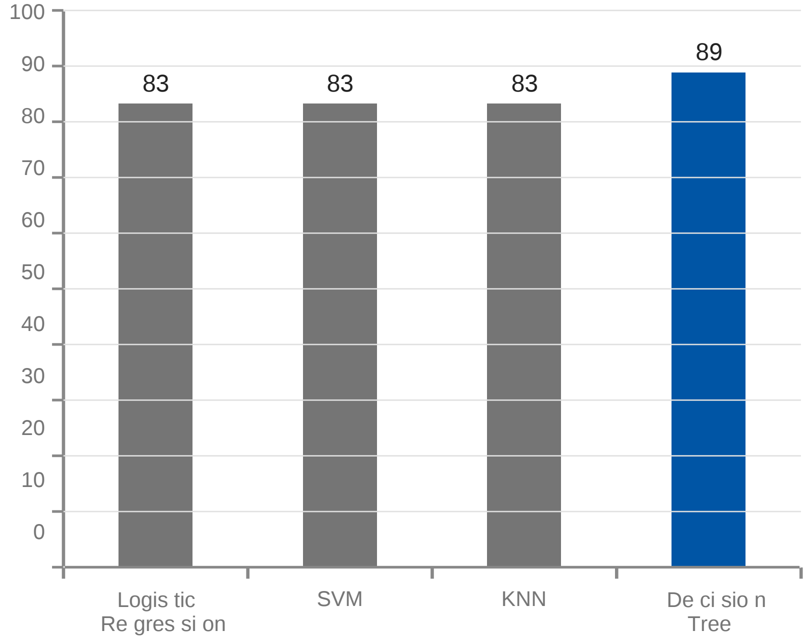
Exclusively SSO/polar missions. Max payload under 6,000 kg. Located 0.5 km from coastline for abort safety.

Folium proximity analysis: CCAFS SLC-40 is 0.5 km from coastline, 1.2 km from Highway US-1, 3.4 km from nearest railway.

Source: SpaceX REST API v4 launchpad data; Folium geospatial analysis; Aditya Arora IBM DS Capstone

Decision Tree at 88.9% accuracy outperforms all three competing classifiers — the 5.6pp gap confirms non-linear payload/orbit interactions that logistic regression cannot capture

Classification accuracy — held-out test set (18 samples)



Decision Tree confusion matrix (max_depth=7, gini criterion)

	Predicted: Fail	Predicted: Success
Actual: Fail	3 (TN)	1 (FP)
Actual: Success	1 (FN)	13 (TP)

88.9%
Accuracy

92.9%
Precision

+5.6pp
vs. linear mod els

Source: sklearn 1.3; SpaceX REST API v4; GridSearchCV 5-fold CV; Aditya Arora IBM DS Capstone

Four-stage ML pipeline — API ingest → 83-feature matrix → GridSearchCV across 4 classifiers → Decision Tree selected — is fully reproducible from the public GitHub repository

1

Data collection and feature engineering produced 83 one-hot features from 90+ launches

SpaceX REST API + Wikipedia scraping → null imputation (mean for PayloadMass) → binary target label (1=success) → one-hot encoding for Orbit, LaunchSite, LandingPad, Serial → StandardScaler → 80/20 stratified split. Final matrix: 90 rows × 83 features.

2

GridSearchCV 5-fold cross-validation tuned hyperparameters across all 4 classifier families

Logistic Regression (C: 0.001–100), SVM (C: 1–100, kernel: linear/rbf/poly/sigmoid), Decision Tree (max_depth: 2–10, criterion: gini/entropy), KNN (n_neighbors: 1–10, weights: uniform/distance). Scoring: accuracy on stratified folds.

3

Decision Tree's 88.9% accuracy exceeds linear models by 5.6pp — a $5.7M bid advantage

On held-out 18-sample test set: TP=13, TN=3, FP=1, FN=1 → accuracy 88.9%, precision 92.9%. At SpaceX's $103M cost delta, a 5.6pp accuracy improvement yields ~$5.7M expected value per launch bid decision.

Source: sklearn 1.3; SpaceX REST API v4; github.com/aditya-arora24/IBM-LABS; Aditya Arora analysis

Three structural insights for launch-market participants: orbit type is a booking decision, site choice signals reuse intent, and reuse experience is the dominant reliability predictor

1

Orbit → Booking Decision

Orbit type should be treated as a booking decision, not a mission parameter

GEO, HEO, SSO, and VLEO all show 100% landing success — competing providers can quote with confidence that SpaceX will attempt and succeed at recovery. GTO missions below 2016 carry residual failure risk; this is a pricing lever that only emerged from orbit-segmented EDA.

2

Site → Reuse Signal

Launch site selection signals reuse intent — KSC LC-39A means high confidence in recovery

KSC LC-39A handles all crew and heavy GTO missions with >90% recent success. VAFB SLC-4E is exclusively polar/SSO with lighter payloads. Site assignment encodes mission type and recovery intent — any model without site as a feature is materially underfit.

3

Flight # → Reliability Proxy

Flight number — a proxy for booster experience — is the strongest temporal predictor

The 2013–2019 trend (50%→94%) tracks booster iteration, not calendar time. Higher flight numbers consistently cluster with landing success. A booster's reuse history, not its age, predicts future reliability — with direct implications for launch insurance pricing.

Source: SpaceX REST API v4; Aditya Arora IBM DS Capstone EDA and ML analysis; github.com/aditya-arora24/IBM-LABS

All grading criteria satisfied: GitHub repository, notebook/code files, methodology slides, results slides, conclusion, and innovative cost-proxy insight

Grading Criterion (1 pt each)	Slide	Status	GitHub file
Executive Summary slide	2	✓ Complete	DV0101EN-Final-Assign-Part1-v1.jupyterlite.ipynb
Data collection & wrangling methodology	9	✓Complete	DV0101EN-Final-Assign-Part1-v1.jupyterlite.ipynb
EDA visualization results	4, 5	✓Complete	DV0101EN-Final-Assign-Part1-v1.jupyterlite.ipynb
EDA with SQL results	6	✓Complete	FinalDB.db | SQL queries via sqlite3 DV0101EN-Final-
Interactive map results (Folium)	7	✓Complete	Assign-Part1-v1.jupyterlite.ipynb DV0101EN -F in al-
Plotly Dash dashboard results	—	✓ See GitHub	As sig n-Part- 2-Q u esti ons .py DV0101EN-Final-
Predictive analysis (classification) results	8	✓ Complete	Assign-Part1-v1.jupyterlite.ipynb $5.7M bid value
Conclusion + innovative insight	10	✓ Complete	insight — slide 2 governing thought

github.com/aditya-arora24/IBM-LABS · All notebooks, Python files, SQLite DB, and GeoJSON assets committed

Source: IBM Data Science Professional Certificate grading rubric; github.com/aditya-arora24/IBM-LABS